

Online Bookstore Database Project

Name: Pooja Bade

SQL Project | Data Analyst

Tools Used: PostgreSQL

Problem Statement

The objective of this project is to design and manage an Online Bookstore database that stores information about books, customers, and orders, and to perform meaningful data analysis using SQL queries.

Database Overview

Table Name	Description
Books	Stores book details such as title, author, genre, price, and stock
Customers	Stores customer personal and location details
Orders	Stores order transactions linking customers and books

SQL Concepts Used

- Database and table creation
- Primary and Foreign Keys
- CRUD operations
- Filtering using WHERE
- Sorting using ORDER BY
- Aggregate functions (SUM, AVG, COUNT)
- GROUP BY and HAVING
- JOINS (INNER, LEFT)
- Subqueries

Queries

-- Create Database

CREATE DATABASE OnlineBookstore;

-- Create Tables

DROP TABLE IF EXISTS Books;

```
CREATE TABLE Books (  
    Book_ID SERIAL PRIMARY KEY,  
    Title VARCHAR(100),  
    Author VARCHAR(100),  
    Genre VARCHAR(50),  
    Published_Year INT,  
    Price NUMERIC(10, 2),  
    Stock INT  
);
```

DROP TABLE IF EXISTS customers;

```
CREATE TABLE Customers (  
    Customer_ID SERIAL PRIMARY KEY,  
    Name VARCHAR(100),  
    Email VARCHAR(100),  
    Phone VARCHAR(15),  
    City VARCHAR(50),  
    Country VARCHAR(150)  
);
```

DROP TABLE IF EXISTS orders;

```
CREATE TABLE Orders (  
    Order_ID SERIAL PRIMARY KEY,  
    Customer_ID INT REFERENCES Customers(Customer_ID),  
    Book_ID INT REFERENCES Books(Book_ID),  
    Order_Date DATE,  
    Quantity INT,  
    Total_Amount NUMERIC(10, 2)  
);
```

SELECT * FROM Books;

	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	1	Configurable modular throughput	Joseph Crane	Biography	1949	21.34	100
2	2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.80	19
3	3	Streamlined coherent initiative	Derrick Howard	Non-Fiction	1913	15.75	27
4	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
5	5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16

SELECT * FROM Customers;

	customer_id [PK] integer	name character varying (100)	email character varying (100)	phone character varying (15)	city character varying (50)	country character varying (150)
1	1	Deborah Griffith	balljoseph@wright-keith.net	1234567891	South Craigfort	Denmark
2	2	Crystal Clements	kimberlybennett@curtis.com	1234567892	East Derekberg	Nicaragua
3	3	Susan Fuller	beanmichael@burnett-stewart.net	1234567893	Austinbury	Equatorial Guinea
4	4	Jamie Ramirez	amandahood@warren.com	1234567894	Dianamouth	Slovenia
5	5	Marcus Murphy	connerjohn@yahoo.com	1234567895	Smithbury	Guinea-Bissau

SELECT * FROM Orders;

	order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	1	84	169	2023-05-26	8	188.56
2	2	137	301	2023-01-23	10	216.60
3	3	216	261	2024-05-27	6	85.50
4	4	433	343	2023-11-25	7	301.21
5	5	14	431	2023-07-26	7	136.36

-- Import Data into Books Table

COPY Books(Book_ID, Title, Author, Genre, Published_Year, Price, Stock)

FROM 'C:/All Excel Practice Files/Books.csv'

CSV HEADER;

-- Import Data into Customers Table

COPY Customers(Customer_ID, Name, Email, Phone, City, Country)

FROM 'C:/All Excel Practice Files/Customers.csv'

CSV HEADER;

-- Import Data into Orders Table

COPY Orders(Order_ID, Customer_ID, Book_ID, Order_Date, Quantity, Total_Amount)

FROM 'C:/All Excel Practice Files/Orders.csv'

CSV HEADER;

- **Retrieve all books from the Fiction genre**

Select * from books

where genre='Fiction';

	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
2	22	Multi-layered optimizing migration	Wesley Escobar	Fiction	1908	39.23	78
3	28	Expanded analyzing portal	Lisa Coffey	Fiction	1941	37.51	79
4	29	Quality-focused multi-tasking challenge	Katrina Underwood	Fiction	1905	31.12	100
5	31	Implemented encompassing conglomerat...	Melissa Taylor	Fiction	2010	21.23	44
6	39	Optimized national process improvement	Megan Goodwin	Fiction	1978	10.99	42
7	40	Adaptive didactic interface	Natalie Gonzalez	Fiction	1923	25.97	94
8	47	Reverse-engineered directional conglome...	John Christian	Fiction	2006	20.37	90
9	62	Re-contextualized real-time strategy	Nicole Lynch	Fiction	1953	26.34	23
10	63	Polarized heuristic database	Franklin Mack	Fiction	1989	22.38	56
11	100	Synchronized client-server service-desk	James Alvarado	Fiction	1906	49.89	29
12	116	Multi-tiered foreground contingency	Jamie Gates	Fiction	1938	41.82	50

- **Find books published after the year 1950:**

Select * from books

where published_year > 1950;

	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.80	19
2	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
3	5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
4	6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.56	2
5	8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.99	84

- **List all customers from the Canada:**

Select * from customers

where country='Canada';

	customer_id [PK] integer	name character varying (100)	email character varying (100)	phone character varying (15)	city character varying (50)	country character varying (150)
1	38	Nicholas Harris	christine93@perkins.com	1234567928	Davistown	Canada
2	415	James Ramirez	robert54@hall.com	1234568305	Maxwelltown	Canada
3	468	David Hart	stokesrebecca@gmail.c...	1234568358	Thompsonfurt	Canada

• **Show orders placed in November 2023:**

Select * from orders
where order_date Between '2023-11-01' And '2023-11-30';

	order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	4	433	343	2023-11-25	7	301.21
2	19	496	60	2023-11-17	9	316.26
3	75	291	375	2023-11-30	5	170.75
4	132	469	333	2023-11-22	7	194.32
5	137	474	471	2023-11-25	8	363.04

• **Retrieve the total stock of books available:**

Select SUM(stock) As Total_Stock
From Books;

	total_stock bigint
1	25056

• **Find the details of the most expensive book:**

select * from Books
Order By Price Desc
Limit 1;

	book_id integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	340	Proactive system-worthy orchestrati...	Robert Scott	Mystery	1907	49.98	88

• **Show all customers who ordered more than 1 quantity of a book:**

select * from orders
where quantity >1;

	order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	1	84	169	2023-05-26	8	188.56
2	2	137	301	2023-01-23	10	216.60
3	3	216	261	2024-05-27	6	85.50
4	4	433	343	2023-11-25	7	301.21
5	5	14	431	2023-07-26	7	136.36

• **Retrieve all orders where the total amount exceeds \$20:**

Select * from Orders
where total_amount>20;

	order_id [PK] integer	customer_id integer	book_id integer	order_date date	quantity integer	total_amount numeric (10,2)
1	1	84	169	2023-05-26	8	188.56
2	2	137	301	2023-01-23	10	216.60
3	3	216	261	2024-05-27	6	85.50
4	4	433	343	2023-11-25	7	301.21
5	5	14	431	2023-07-26	7	136.36

- **List all genres available in the Books table:**

Select Distinct genre from Books;

	genre character varying (50) 🔒
1	Romance
2	Biography
3	Mystery
4	Fantasy
5	Fiction
6	Non-Fiction
7	Science Fiction

- **Find the book with the lowest stock:**

Select * from books

order by stock

Limit 1;

	book_id [PK] integer 🔒	title character varying (100) 🔒	author character varying (100) 🔒	genre character varying (50) 🔒	published_year integer 🔒	price numeric (10,2) 🔒	stock integer 🔒
1	44	Networked systemic implementati...	Ryan Frank	Science Fiction	1965	13.55	0

- **Calculate the total revenue generated from all orders:**

Select SUM(total_amount) as Revenue

From orders;

	revenue numeric 🔒
1	75628.66

- **Retrieve the total number of books sold for each genre:**

select * from orders;

Select b.genre,SUM(o.quantity) As Total_Books_Sold

From Orders o

Join Books b On o.book_id=b.book_id

Group by b.genre;

	genre character varying (50) 🔒	total_books_sold bigint 🔒
1	Romance	439
2	Biography	285
3	Mystery	504
4	Fantasy	446
5	Fiction	225
6	Non-Fiction	351
7	Science Fiction	447

- Find the average price of books in the "Fantasy" genre:

Select AVG(price) As Average_price
From Books
Where genre='Fantasy';

	average_price numeric
1	25.9816901408450704

- List customers who have placed at least 2 orders:

Select o.customer_id,c.name,Count(o.order_id) As Order_count
from orders o
Join customers c on o.customer_id=c.customer_id
Group By o.customer_id ,c.name
having count(order_id) >= 2;

	customer_id integer	name character varying (100)	order_count bigint
1	225	Christopher Mccullough	2
2	418	Kiara Blankenship MD	3
3	322	William Cameron	3
4	325	Emily Vargas	4
5	376	Justin Donaldson	2

- Find the most frequently ordered book:

select o.book_id,b.title,Count(o.order_id) As Order_count
from orders o
Join Books b On o.book_id=b.book_id
Group By o.book_id,b.title
order by order_count Desc Limit 1;

	book_id integer	title character varying (100)	order_count bigint
1	88	Robust tangible hardwa...	4

- Show the top 3 most expensive books of 'Fantasy' Genre :

select * from books
where genre='Fantasy'
order by price desc
limit 3;

	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (50)	published_year integer	price numeric (10,2)	stock integer
1	240	Stand-alone content-based hub	Lisa Ellis	Fantasy	1957	49.90	41
2	462	Innovative 3rdgeneration databa...	Allison Contreras	Fantasy	1988	49.23	62
3	238	Optimized even-keeled analyzer	Sherri Griffith	Fantasy	1975	48.97	72

- **Retrieve the total quantity of books sold by each author:**

```
select b.author,SUM(o.quantity) As Total_Books_Sold
from Orders o
Join books b on o.book_id=b.book_id
Group by b.Author;
```

	author character varying (100) 🔒	total_books_sold bigint 🔒
1	Jared Cortez	10
2	Tracy Parker	11
3	Taylor Wang	9
4	Cathy Knight	6
5	Bianca Matthews	3

- **List the cities where customers who spent over \$30 are located:**

```
select Distinct c.city ,total_amount
from orders o
Join customers c On o.customer_id=c.customer_id
where o.total_amount > 30;
```

	city character varying (50) 🔒	total_amount numeric (10,2) 🔒
1	Taylorfurt	189.45
2	Leeport	141.39
3	Port Jasonview	149.12
4	Port Aaronstad	145.44
5	Matthewfurt	328.50

- **Find the customer who spent the most on orders:**

```
select c.customer_id,c.name,Sum(o.total_amount) As Total_Spent
From orders o
Join customers c On o.customer_id=c.customer_id
Group By c.customer_id,c.name
order By Total_spent desc Limit 1;
```

	customer_id [PK] integer 🔒	name character varying (100) 🔒	total_spent numeric 🔒
1	457	Kim Turner	1398.90

- **Calculate the stock remaining after fulfilling all orders:**

```
select b.book_id,b.title,b.stock,Coalesce(Sum(quantity),0) As Order_quantity,
       b.stock- Coalesce(Sum(o.quantity),0) As Remaining_quantity
```

From books b

Left join orders o On b.book_id=o.book_id

Group By b.book_id order by b.book_id;

	book_id [PK] integer	title character varying (100)	stock integer	order_quantity bigint	remaining_quantity bigint
1	1	Configurable modular throughput	100	3	97
2	2	Persevering reciprocal knowledge user	19	0	19
3	3	Streamlined coherent initiative	27	5	22
4	4	Customizable 24hour product	8	0	8
5	5	Adaptive 5thgeneration encoding	16	8	8

Key Learnings & Insights

Through this project, I learned how to design a relational database, write optimized SQL queries, and extract meaningful business insights such as revenue, customer behaviour, and inventory status.

Conclusion

This project demonstrates practical SQL skills required for a Data Analyst role and showcases the ability to work with real-world relational databases.